CSAT Hard Revision Folder 2: Number systems, arithmetic, rates, geometry and table-based data interpretation

ANSWER KEY + SOLUTIONS | 40 questions | Source: original synthesis

Scoring: +2.5 for correct, –5/6 for incorrect, 0 for blank. Check the key only after completing the answer-free paper.

Q	Key	Category
1	A	Comprehension
2	A	Comprehension
3	B	Comprehension
4	B	Comprehension
5	A	Comprehension
6	D	Comprehension
7	B	Comprehension
8	B	Comprehension
9	B	Comprehension
10	B	Comprehension
11	D	Numeracy / DI
12	B	Numeracy / DI
13	C	Numeracy / DI
14	C	Numeracy / DI
15	A	Numeracy / DI
16	A	Numeracy / DI
17	A	Numeracy / DI
18	B	Numeracy / DI
19	C	Numeracy / DI
20	D	Numeracy / DI
21	B	Logical reasoning
22	C	Logical reasoning
23	B	Logical reasoning
24	D	Logical reasoning
25	B	Logical reasoning
26	B	Logical reasoning
27	A	Logical reasoning
28	B	Logical reasoning
29	C	Logical reasoning
30	D	Logical reasoning
31	B	Decision-making / problem solving
32	C	Decision-making / problem solving
33	D	Decision-making / problem solving
34	B	Decision-making / problem solving

Q	Key	Category
35	D	Decision-making / problem solving
36	D	Decision-making / problem solving
37	D	Decision-making / problem solving
38	D	Decision-making / problem solving
39	A	Decision-making / problem solving
40	B	Decision-making / problem solving

DETAILED SOLUTIONS

Q1 — A Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q2 — A Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q3 — B Comprehension
The argument depends on learning through feedback; it does not require perfect information or reject all central rules.
Provenance: Original synthesis

Q4 — B Comprehension
The wording weighs trade-offs and improvement mechanisms rather than celebrating, rejecting or attacking them.
Provenance: Original synthesis

Q5 — A Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q6 — D Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q7 — B Comprehension
The argument depends on learning through feedback; it does not require perfect information or reject all central rules.
Provenance: Original synthesis

Q8 — B Comprehension
The wording weighs trade-offs and improvement mechanisms rather than celebrating, rejecting or attacking them.
Provenance: Original synthesis

Q9 — B Comprehension
The passage links the issue to incentives, evidence and feedback; the alternatives are absolute or narrowly framed.
Provenance: Original synthesis

Q10 — B Comprehension
The passage distinguishes a desirable objective from the quality of its feedback loop; it rejects universal claims.
Provenance: Original synthesis

Q11 — D Numeracy / DI
Compute powers modulo 13; cyclic reduction gives remainder 2.
Provenance: Original synthesis

Q12 — B Numeracy / DI
Count multiples of 7 and 14, then remove twice the common multiples: the exclusive count is 8.
Provenance: Original synthesis

Q13 — C Numeracy / DI
Use a weighted average: (5×29+13)/(5+1) = 26.33%.
Provenance: Original synthesis

Q14 — C Numeracy / DI
Combined rate = 1/18+1/20+1/32; reciprocal = 7.31 days.
Provenance: Original synthesis

Q15 — A Numeracy / DI
Change in average = (new−old)/25; new average = 56.08.
Provenance: Original synthesis

Q16 — A Numeracy / DI
Multiply successive factors: 1.0×(1+17/100)×(1−23/100)×(1+29/100); net = 16.22%.
Provenance: Original synthesis

Q17 — A Numeracy / DI
Removed area = 4×2^2; remaining area = $12\times10-4\times2^2 = 104$ m².
Provenance: Original synthesis

Q18 — B Numeracy / DI
Weighted total = 126; 126 divided by 17 leaves remainder 7.
Provenance: Original synthesis

Q19 — C Numeracy / DI
lcm(9,10) = 90; enumerate its multiples in the interval to obtain 3.
Provenance: Original synthesis

Q20 — D Numeracy / DI
The increments alternate 6, 5, 6, 5; the next increment is 6, giving 33.
Provenance: Original synthesis

Q21 — B Logical reasoning
Enumerating all valid arrangements gives possible fifth-position occupants ['A', 'B', 'D', 'E', 'F', 'G']; C never appears there.
Provenance: Original synthesis

Q22 — C Logical reasoning
Testing all rankings gives possible third-place candidates ['P', 'Q', 'R', 'S', 'T']; U violates the partial-order constraints in every valid ranking.
Provenance: Original synthesis

Q23 — B Logical reasoning
(7×6+3) mod 26 = 19.
Provenance: Original synthesis

Q24 — D Logical reasoning
Y contains X plus Y-only members: 5+5 = 10.
Provenance: Original synthesis

Q25 — B Logical reasoning
Displacement = sqrt(7^2+5^2) = 8.60 km.
Provenance: Original synthesis

Q26 — B Logical reasoning
N is P's father; M is N's sister, so M is P's paternal aunt.
Provenance: Original synthesis

Q27 — A Logical reasoning
Assume exactly one of A, B and C is guilty and exactly one statement is true. A says B is guilty; B says A and C are innocent; C says B is innocent. Testing A, B and C in turn leaves only B guilty: A and C are then false, while B is true.
Provenance: Original synthesis

Q28 — B Logical reasoning
All selections minus those containing both specified proposals: C(11,4)−C(9,2) = 294.
Provenance: Original synthesis

Q29 — C Logical reasoning
Points = 3×5+4 = 19; losses add zero.
Provenance: Original synthesis

Q30 — D Logical reasoning
The first post has 6 choices and the second has 5; ordered pairs therefore total 6×5 = 30.
Provenance: Original synthesis

Q31 — B Decision-making / problem solving
Disaggregated evidence supports adaptation rather than premature scaling, cancellation or concealment.
Provenance: Original synthesis

Q32 — C Decision-making / problem solving
Complaint counts need accessibility and independent-feedback checks before they can indicate improvement.
Provenance: Original synthesis

Q33 — D Decision-making / problem solving
A documented risk and change-control assessment protects safety without making procedure an empty ritual.
Provenance: Original synthesis

Q34 — B Decision-making / problem solving
Transparent deduplication plus appeal and verification addresses both leakage and wrongful exclusion.
Provenance: Original synthesis

Q35 — D Decision-making / problem solving
Equal application of a published rule protects fairness while allowing genuinely defined hardship relief.
Provenance: Original synthesis

Q36 — D Decision-making / problem solving
The response preserves useful innovation while removing exclusion through assisted access, fallback and review.
Provenance: Original synthesis

Q37 — D Decision-making / problem solving
A shared evidence base and explicit assumptions address disagreement without arbitrary choice or paralysis.
Provenance: Original synthesis

Q38 — D Decision-making / problem solving
Outcome-by-issue plus verified follow-up distinguishes visible activity from effective service delivery.
Provenance: Original synthesis

Q39 — A Decision-making / problem solving
The evidence-supported intervention creates an accountable learning loop and a reviewable use of funds.
Provenance: Original synthesis

Q40 — B Decision-making / problem solving
Consistent use of a pre-published rule protects equal treatment, transparency and auditability.
Provenance: Original synthesis